Saumya Vijayvargiya
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Education

- **BITS Pilani K K Birla Goa Campus** Goa, India
  Master of Science - Biological Sciences *August 2024 - August 2029*

  Bachelor of Engineering - Electronics and Communication Engineering CGPA: 8.62

Projects

- **Aqueous Solubility Prediction** Advised by Prof. Raviprasad Aduri
  Computational Biology & Drug Delivery | Python, PyTorch, RDKit
  - **Model Architecture:** Developed a Message Passing Neural Network (MPNN) to predict aqueous solubility of drug-like molecules directly from SMILES string representations, leveraging graph-based molecular encoding.
  - **Chemical Space Analysis:** Analyzed the chemical space of SMILES-encoded molecules to extract structural and physicochemical insights, enabling a deeper understanding of molecular properties influencing solubility.
  - **Data Pipeline:** Engineered a robust data pipeline with an 80/20 train-test split and 5-fold cross-validation on the training set to ensure model generalizability and prevent overfitting.
  - **Feature Engineering:** Applied feature importance techniques to identify and select the top chemical descriptors, optimizing model input and improving predictive performance.
  - **Benchmarking:** Benchmarked MPNN performance against classical ML models including Random Forest, LightGBM, and XGBoost, demonstrating the effectiveness of graph-based deep learning for molecular property prediction.
- **Relevant Courses:**
  Machine Learning Specialization — Andrew Ng (Coursera)
  Global Consumer Intelligence (GCI) course by University of Tokyo
  Bioinformatics | Nanobiotechnology | Genetics | Cell Biology | Microbiology

Extracurricular

- **Formula Bharat** National Level Competition
  Core Powertrain Engineer in Team GigaWatt *2024 - Present*
  Powertrain System Design: Designed and integrated mechanical powertrain systems from scratch using CAD tools for a Formula-style race vehicle.
  Cross-functional Collaboration: Worked within a high-performance, multidisciplinary team managing tight deadlines and complex system dependencies under competitive pressure.
  Systems Integration: Coordinated powertrain components with chassis and electrical subsystems, ensuring reliability and performance under race conditions.
- **Department of Creative Works** College Fest Organization
  Core Member
  - Actively involved in planning, coordinating, and executing events during college fests as part of the creative works department.

Skills

- **Languages** Python, C, Java
- **ML** Graph-based Neural Networks, Chemical Space Exploration, Regression Modeling, Cheminformatics
- **Frameworks** PyTorch, TensorFlow, Scikit-learn, RDKit
- **Tools** SOLIDWORKS, Fusion 360, Jupyter Notebook, Google Colab